Capstone Project: ML-Driven Decision Support for "Aurora Finance"

Scenario:
Aurora Finance is a fictitious mid-size financial services firm that does corporate lending, manages investments, and trades derivatives. They want to leverage ML for better decision-making across the firm:

- **Which corporate projects to fund**
- **How to price loans and assess credit risk**
- **Where to invest and construct portfolios**
- **How to monitor and hedge derivative exposures**

Students will act as "Data & ML Strategy Consultants", integrating ML insights into financial decision-making.

Step 0: Project Setup

- **Team: 4–5 students (recommended)**
- **Timeline: Weeks 1–14**
- **Deliverables:**
    1. **Executive-style project brief**
    2. **Data exploration & model outputs (notebook or spreadsheet)**
    3. **Dashboard / visualization summarizing insights**
    4. **Final presentation/report (10–15 slides)**
- **Key objectives: Combine predictive modeling, decision-making metrics, and interpretability.**

Step 1: Corporate Finance Module

Objective: Decide which internal projects Aurora should fund.

Data Provided / Simulated:

- **Historical corporate projects: investment amount, expected cash flows, duration, risk rating, department, past ROI**
- **Market indicators: GDP growth, sector-specific trends**

Tasks:

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1. **Forecast project cash flows using regression / ML (e.g., XGBoost / Random Forest)**
2. **Predict project success probability / IRR using classification or risk scoring**
3. **Recommend top projects based on expected economic value (EV) and risk-adjusted return**

Deliverables:

- **Ranked project list**
- **SHAP plots / decision explanation for executives**
- **Brief summary of recommendations**

Step 2: Banking Module

Objective: Assess loan portfolio and detect fraud.

Data Provided / Simulated:

- **Corporate and retail loan applications: income, balance sheet ratios, past defaults, loan type, duration**
- **Transactional dataset: anomalies (fraud, late payments)**

Tasks:

1. **Credit Risk: Predict probability of default (PD) and classify high-risk loans**
2. **Fraud Detection: Flag anomalous transactions using supervised or unsupervised methods**
3. **Recommend loan approval or monitoring strategy**

Deliverables:

- **Risk-scored loan portfolio**
- **Fraud alerts / visual dashboard**
- **Short summary for lending officers**

Step 3: Financial Markets Module

Objective: Suggest investment strategies for Aurora's portfolio.

Data Provided / Simulated:

- **Historical stock prices, macroeconomic indicators, sector indices**

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- **Historical portfolio allocations, returns, risk metrics**

Tasks:

1. **Predict next-period stock returns using ML (tree-based or regression models)**
2. **Build optimized portfolio using predicted returns + risk constraints**
3. **Backtest strategy over last 2 years (simple moving window backtesting)**

Deliverables:

- **Predicted return table / ranking**
- **Recommended portfolio allocation**
- **Backtest summary: expected vs realized returns**
- **Executive slide: "ML insights for investment decisions"**

Step 4: Derivatives Module

Objective: Hedge portfolio risk and price derivative exposure.

Data Provided / Simulated:

- **Option contracts: strikes, maturities, underlying asset prices, historical volatility**
- **Portfolio from Step 3**

Tasks:

1. **Predict option prices or implied volatilities using ML**
2. **Suggest hedging strategy for portfolio (delta/gamma hedging simplified)**
3. **Evaluate risk exposure metrics (portfolio VaR)**

Deliverables:

- **Hedged portfolio recommendations**
- **Visuals of derivative pricing predictions**
- **Risk dashboard**

Step 5: Integrated Executive Decision Dashboard

Combine outputs from all modules into a single, decision-focused dashboard:

- **Project funding recommendations**

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- Loan portfolio & fraud alerts
- Investment portfolio allocation & backtest
- Derivatives risk & hedging recommendations

Deliverables:

- Executive slide deck (~15 slides)
- Decision dashboards (Excel / Tableau / PowerBI / Python interactive)
- Notebook with reproducible ML workflows

Step 6: Executive Summary / Presentation

- 1–2 page summary of all ML-driven recommendations for senior management
- Include key metrics, top 3 action items per function, and risk considerations
- Highlight interpretability & business reasoning, not just ML metrics

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Step 7: Evaluation Criteria

Criteria	Weight
ML Model Accuracy & Insights	25%
Business Decision Alignment	25%
Integration Across Modules	20%
Explainability	15%
Executive Communication	15%

1⃣ Corporate Finance Module — Project Funding

Dataset: corporate_projects.csv

Column	Type	Description
Project_ID	int	Unique identifier

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Column	Type	Description
Department	str	Finance, IT, Operations, Marketing
Investment_Cost	float	$ amount required
Expected_Cashflow_Year1	float	Estimated cash flow year 1
Expected_Cashflow_Year2	float	Estimated cash flow year 2
Expected_Cashflow_Year3	float	Estimated cash flow year 3
Historical_ROI	float	ROI of past projects (similar type)
Market_Growth	float	% industry growth
Project_Risk	str	Low, Medium, High
Success	int	1=successful project, 0=failed (for supervised ML)

Notebook Tasks:

- Regression to predict NPV / cash flows
- Classification for project success
- Ranking projects by expected EV / risk-adjusted return
- SHAP plots for executive interpretation

2 Banking Module — Credit Risk & Fraud

Datasets:

1. loan_portfolio.csv

Column	Type	Description
Loan_ID	int	Unique loan ID
Customer_Type	str	Corporate / Retail
Annual_Income	float	$
Debt_to_Income	float	Ratio
Credit_History_Length	int	Years
Loan_Amount	float	$

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Column	Type	Description
Interest_Rate	float	%
Loan_Term_Months	int	Duration
Past_Default	int	0/1
PD	float	Target variable for supervised ML

2. transactions.csv

Column	Type	Description
Transaction_ID	int	Unique ID
Customer_ID	int	Link to loan/customer
Amount	float	$
Transaction_Type	str	Payment, Withdrawal, Transfer
Timestamp	datetime	Time of transaction
Fraud_Flag	int	0/1

Notebook Tasks:

- Predict PD (credit risk) using ML classifiers
- Fraud detection with supervised or anomaly detection methods
- Executive-friendly risk visualization

3⃣ Financial Markets Module — Investment Strategy

Dataset: market_data.csv

Column	Type	Description
Ticker	str	Stock symbol
Date	date	Daily observation
Open	float	Opening price
Close	float	Closing price

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Column	Type	Description
High	float	High price
Low	float	Low price
Volume	int	Trading volume
Sector	str	Industry sector
Macro_GDP_Growth	float	% quarterly GDP growth
Macro_Interest_Rate	float	%
Return	float	Daily return (target)

Notebook Tasks:

- Predict next-day / next-week returns using ML models
- Rank stocks for investment
- Portfolio optimization (Markowitz or simplified)
- Backtesting returns
- Visualizations for executives

4⃣ Derivatives Module — Option Pricing / Hedging

Dataset: options_data.csv

Column	Type	Description
Option_ID	int	Unique identifier
Underlying_Ticker	str	Stock
Strike	float	Strike price
Expiry	date	Expiration date
Option_Type	str	Call / Put
Underlying_Price	float	Current stock price
Historical_Volatility	float	%

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Column	Type	Description
Interest_Rate	float	% risk-free rate
Market_Price	float	Observed option price (for supervised ML)

Notebook Tasks:

- **Predict option prices using ML regression**
- **Compare to Black–Scholes theoretical price**
- **Suggest hedging strategy (simplified delta / gamma)**
- **Risk dashboard visualization**

5⃣ Integrated Executive Dashboard

Notebook Components:

- **Combine outputs from all 4 modules into one decision dashboard**
- **Executive view:**
  - **Top projects to fund**
  - **High-risk loans & fraud alerts**
  - **Recommended portfolio allocation & expected return**
  - **Derivative hedging recommendations**
- **Use Plotly / Matplotlib / Seaborn for charts, or Excel/PowerBI for executive-friendly dashboards**

6⃣ Notebook Features for Executives

- **Pre-filled data exploration cells: summary stats, distributions, correlations**
- **ML cells with default models: Random Forest, XGBoost, Logistic Regression**
- **Interpretability plots: SHAP, feature importance**
- **Guidance comments: explain outputs in business terms**
- **Downloadable dashboard snapshot: Excel / CSV for executives**

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